Loan repayments

1 Iain takes out a loan to purchase a jetski. He makes 15 equal loan repayments of $\$5239$. Calculate the total amount paid back on the loan.

2 Dylan takes out a loan to purchase a property. He makes equal monthly loan repayments of $\$4600$ over 27 years to pay it off. Calculate the total amount paid back on the loan.

3 Gwen takes out a loan to purchase a surround sound system. She makes 11 equal loan repayments. The total loan amount paid back is $\$6600$.
Calculate the amount of each repayment.

4 Lisa takes out a loan to purchase a small boat. She pays it back in equal monthly repayments over 6 years. The total loan amount paid back is $\$55\,800$.
Calculate the amount of each repayment.

5 Jimmy takes out a loan of $\$700$ to purchase a computer. The loan earns interest at 6% p.a. compounded annually. Repayments of $\$140$ are made annually.

a Calculate the amount Jimmy still owes after his first repayment.

b Calculate the amount Jimmy still owes after his second repayment.

6 Kathleen takes out a loan of $\$88\,000$ to renovate her home. The loan earns interest at 8% p.a. compounded monthly. Repayments of $\$11\,264$ are made annually.

a Calculate the amount Kathleen will still owe after the first repayment.

b Calculate the amount Kathleen will still owe after the second repayment.

7 Han received a 9-year $\$43\,000$ loan at 10% p.a. monthly reducible interest. He makes monthly instalments of $\$209$.

a Find the amount owing after:

i 1 month ii 2 months iii 3 months

b Is the amount owing increasing or decreasing?

c Will this loan eventually be paid off if the instalments remain the same? Explain your answer.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

8 Joanne takes out a loan of $\$54\,000$ to purchase a plot of land. The loan earns interest at 2% p.a. compounded annually. Repayments of $\$864$ are made annually.

a How much interest is added to Joanne's loan in the first year?

b Calculate the amount Joanne still owes after the first repayment.

c Should Joanne continue to make annual payments of $\$864$? Explain your answer.

d Suggest a better annual payment amount for Joanne's loan.

9 Luke takes out a loan of $\$73\,200$ to start a business. The loan earns interest at 8% p.a. compounded annually. State whether the loan will be paid off in 3 years if the following amounts are repaid monthly:

a $\$1627$ **b** $\$2034$ **c** $\$2562$

10 Valentina is deciding between two $\$109\,000$ home loans. She has the capacity to pay $\$3400$ per month.

- Option 1: 3.2% p.a. over 3 years with fixed monthly repayments of $\$3179$.
- Option 2: 2.6% p.a. over 3 years with minimum monthly repayments of $\$3151$ that enables paying more than the minimum monthly repayment

a What is the total repayment she will have to make with Option 1?

b What is the maximum total repayment she will have to make with Option 2?

c Which loan will cost less?

Reducing balance loan tables

11 Ivan takes out a car loan for $\$24\,000$. He is charged 8.1% per annum interest, compounded monthly. Ivan makes repayments of $\$450$ at the end of each month.
Complete the following table which tracks the loan over the first three months:

Month	Opening Balance	Interest	Repayment	Closing Balance
1	24 000	162	450	23 712
2				
3				

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

12 A car loan of $\$6000$ earns interest at 8% p.a. compounded annually. Repayments of $\$720$ are made annually.
Complete the following table which tracks the loan over the first three years:

Time period (n)	Value at beginning of time period	Interest at end of time period	Repayment this period	Amount at end of time period
1	$\$6000$			
2		$\$460.80$	$\$720$	
3	$\$5500.80$		$\$720$	

13 A loan of $\$54\,000$ earns interest at 5.4% p.a. compounded monthly. Repayments of $\$1500$ are made monthly.
Complete the following table which tracks the loan amount over the first three months:

Time period (n)	Value at beginning of time period	Interest at end of time period	Repayment this period	Amount at end of time period
1		$\$243.00$	$\$1500$	$\$52\,743$
2		$\$237.34$		
3	$\$51\,480.34$	$\$231.66$	$\$1500$	

14 Dave took out a loan of $\$6500$ to start his business on January 1. Interest on the loan is charged at 10% p.a. from the time the loan is taken out. He made yearly repayments of $\$2500$ on December 31 each year.

a Complete the repayment table:

Year	Balance owing at beginning of year	Interest charged in this period	Balance owing at end of year	Balance owing after repayment
1				
2				
3				

b Find the amount of interest charged between his first and second repayment.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

15 Monthly repayments of $\$3990$ are made on a loan of $\$158\,800$ borrowed at a rate of 12% p.a. compounded monthly.
Consider the following table of home loan repayments:

Month (n)	Principal (P)	Interest (I)	$P+I$	$P+I-R$
1	$\$158\,800$	$\$1588.00$	$\$160\,388.00$	$\$156\,398.00$
2	$\$156\,398.00$	$\$1563.98$	$\$157\,961.98$	$\$153\,971.98$
3				

a What is the principal at the beginning of the third month?

b Calculate the interest charged for the third month.

c How much money is owed at the beginning of the fourth month?

16 The following table shows the principal and interest over the first 4 months of a loan:

Month	P	I	$P+I$	$P+I-R$
1	$\$20\,000$	$\$50.00$	$\$20\,050.00$	$\$19\,150.00$
2	$\$19\,150.00$	$\$47.88$	$\$19\,197.88$	$\$18\,297.88$
3	$\$18\,297.88$	$\$45.74$	$\$18\,343.62$	$\$17\,443.62$
4	$\$17\,443.62$	$\$43.61$	$\$17\,487.23$	$\$16\,587.23$

a Find the value of each repayment, R.

b What is the annual interest rate charged on the loan to the nearest percent?

c Calculate the value of the principal at the start of month 6.

17 Kate takes out a personal loan for $\$40\,000$. The interest on the loan is charged quarterly and Kate makes repayments of $\$900$ at the end of each quarter.
Calculations for the first quarter are shown in the following table:

Quarter	Opening Balance	Interest	Repayment	Closing Balance
1	40 000	400	900	39 500
2				
3				

a Calculate the quarterly interest rate Kate is charged for this loan as a percentage.

b State the annual interest rate of this loan as a percentage.

c Calculate the values for Quarters 2 and 3 in the table.

d How much of the loan has Kate repaid in the first three quarters of the year?

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

18 Maximilian takes out a mortgage to purchase an investment property. A number of his payments and balances are shown in the following table:

Month	Opening Balance	Interest	Repayment	Closing Balance
1	x		3500	417 970
2	417 970	1462.90	3500	
3		y		

a Calculate the monthly interest rate charged on this loan as a percentage. Round your answer to two decimal places.

b Calculate the annual interest rate charged on this loan as a percentage. Round your answer to two decimal places.

c Calculate the value of x, the initial amount borrowed for this mortgage.

d Calculate the value of y in the table.

19 A study abroad loan of $8500 earns interest which is compounded monthly. Repayments of $2550 are made half yearly. The following table documents the repayment of the loan:

Time period (n)	Value at beginning of time period	Interest at end of time period	Repayment this period	Amount at end of time period
1	$8500	$416.25	$2550	$6366.25
2	$6366.25	$311.76	$2550	$4128
3	$4128	$202.15	$2550	$1780.15
4	$1780.15	$87.17	$1867.33	$0

Calculate the total loan amount paid.

20 A small loan of $4500 to pay for a holiday earns interest at 4% p.a. compounded annually. Repayments of $2000 are made annually.

a Complete the following table which tracks the loan amount over three years:

Time period (n)	Value at beginning of time period	Interest at end of time period	Repayment this period	Amount at end of time period
1	$4500		$2000	
2		$107.20		
3	$787.20			$0

b Why is the repayment in the third year smaller than the other repayments?

c Calculate the total loan amount paid.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

21 You take out a personal loan of $\$10\,000$ at 11% reducible p.a. The term of the loan is 3 years, and yearly repayments of $\$2600$ are made. The balance owing is paid at the end of 3 years.

a Complete the loan repayment table:

Time period	Balance owing at beginning of period	Balance owing plus interest during period	Repayment	Balance owing at end of period
1	10 000		2600	
2			2600	
3				

b Find the total amount of interest charged on the loan.

c Find the total repayment over 3 years.

22 A study abroad loan of $\$13\,600$ earns interest at 2.4% p.a. compounded monthly. Repayments of $\$4080$ are made either half-yearly or yearly.

a Complete the following table which tracks the repayment of the loan with half-yearly payments:

Time period (n)	Value at beginning of time period	Interest at end of time period	Repayment this period	Amount at end of time period
1	$\$13\,600$	$\$164.02$	$\$4080$	
2	$\$9684.02$	$\$116.79$		
3	$\$5720.81$	$\$68.99$	$\$4080$	
4				$\$0$

b What is the total loan amount paid if making half yearly repayments?

c Complete the following table that tracks the repayment of the loan with yearly payments:

Time period (n)	Value at beginning of time period	Interest at end of time period	Repayment this period	Amount at end of time period
1	$\$13\,600$	$\$330.01$	$\$8160$	
2	$\$5770.01$	$\$140.01$		

d What is the total loan amount paid if making yearly repayments?

e What is the better payment option in order to reduce the total amount paid?

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

23 $3700 is invested for three years at a rate of 7% p.a., compounded annually.

a Complete the table below to determine the final value of the investment:

	Balance at beginning of year	Interest earned
First year	$3700	$259
Second year	$3959	$277.13
Third year		
Fourth year		–

b Calculate the total interest earned over the three years.

24 Mr and Mrs Lyne have a $520 000 mortgage for their home. They are charged 5.16% interest per annum, compounded monthly and make monthly repayments of $3750.

a Complete the table below, using the rounded answer to calculate the amounts for the following month:

Month	Opening Balance ($)	Interest ($)	Repayment ($)	Closing Balance ($)
1	520 000	2236	3750	518 486
2				
3				
4				
5				

b If they made fortnightly repayments of $1 875 instead of monthly repayments, what would be the result on the balance of their loan after 10 years?

25 A $2000 loan is to be repaid at a reducible rate of 8% p.a. There are two possible methods of repayment: repaying $750 every 6 months or repaying $375 every 3 months, and the interest is compounded at the same interval as payment.

a Complete the repayment table, showing repayments made every 6 months:

Time period	Balance owing at beginning of period	Balance owing plus interest during period	Repayment	Balance owing at end of period
1	2000	2080	750	
2			750	

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

b Complete the repayment table, showing repayments made every 3 months:

Time period	Balance owing at beginning of period	Balance owing plus interest during period	Repayment	Balance owing at end of period
1	2000	2040	375	1665
2	1665	1698.30	375	
3			375	
4			375	

c Calculate the difference in the balances owing after 1 year.

26 A credit card charges annual interest of 18.24%.

a Calculate the monthly interest rate on this credit card, correct to two decimal places.

b Roald owes $\$3200$ on his credit card and will pay it all off before using this card again. The table displays his payments:

Month	Interest	Repayment	Amount Owing
			3200
1	48.67	75	X
2	48.27	75	3146.93
3	Y	75	3119.79
$\ldots$	$\ldots$	$\ldots$	$\ldots$
$\ldots$	$\ldots$	$\ldots$	$\ldots$
66	4.76	75	242.66
67	3.69	75	171.35
68	2.61	75	98.96
69	1.51	75	25.47
70	Z	W	0

i Calculate the value of X.

ii Calculate the value of Y

iii Calculate the value of Z.

iv Calculate the value of W.

v Calculate the total interest paid.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

27 Xanthe takes out a car loan. The last few months of her repayments are shown in the following table:

Month	Opening Balance	Interest	Repayment	Closing Balance
69	868.48	4.34	250	622.82
70	622.82	3.11	250	375.93
71	375.93	1.88	250	127.81
72				

a Calculate the monthly interest rate charged on this loan, rounding your answer to one decimal place.

b Calculate the values for the final row of the table, rounding your answers correct to two decimal places.

c How many years did it take for Xanthe to pay off the loan?

d Calculate her total repayments.

e If her original loan was for $\$15\,000$, calculate the total interest paid on the loan.

28 Mr. and Mrs. Dave have a mortgage. The final months of their repayments are shown in the following table:

Month	Opening Balance	Interest	Repayment	Closing Balance
146	26 452.84	198.40	5000	21 651.24
147	21 651.24	162.38	5000	16 813.62
148	16 813.62	126.10	5000	11 939.72
149	11 939.72	89.55	5000	7029.27
150	7029.27	52.72	5000	2081.99
151				

a Calculate the monthly interest rate charged on this loan. Round your answer to two decimal places.

b Calculate the values for the final row of the table, rounding your answers to two decimal places.

c How many years did it take for them to pay off the loan? Round your answer to two decimal places.

d Calculate the total repayments.

e If they paid $\$302\,097.60$ in interest, how much did they initially borrow?

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Financial tables

29 The following financial table displays the monthly repayments on a $1000 loan:

Annual interest rate	5 years	10 years	15 years	20 years	25 years
10%	21.25	13.22	10.75	9.65	9.09
11%	21.74	13.78	11.37	10.32	9.80
12%	22.24	14.35	12.00	11.01	10.53
13%	22.75	14.93	12.65	11.72	11.28
14%	23.27	15.53	13.32	12.44	12.04
15%	23.79	16.13	14.00	13.17	12.81

Han received a 5-year $151 000 loan at 10% p.a. monthly reducible interest. If the interest rate was increased to 15% p.a., find the increase in the amount of each monthly instalment needed to be paid.

30 The following financial table displays the monthly repayments on a $1000 loan:

Annual interest rate	5 years	10 years	15 years	20 years	25 years
6%	19.33	11.10	8.44	7.16	6.44
7%	19.80	11.61	8.99	7.75	7.07
8%	20.28	12.13	9.56	8.36	7.72
9%	20.76	12.67	10.14	9.00	8.39
10%	21.25	13.22	10.75	9.65	9.09
11%	21.74	13.78	11.37	10.32	9.80

Maria received a 5-year $209 000 loan at 10% p.a. monthly reducible interest. If the interest rate fell to 6% p.a., find the decrease in the amount of each monthly instalment needed to be paid.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

31 The following financial table displays the monthly repayments on a $1000 loan:

Annual interest rate	10 years	15 years	20 years	25 years	30 years
4%	10.12	7.40	6.06	5.28	4.77
5%	10.61	7.91	6.60	5.85	5.37
6%	11.10	8.44	7.16	6.44	6.00
7%	11.61	8.99	7.75	7.07	6.65
8%	12.13	9.56	8.36	7.72	7.34
9%	12.67	10.14	9.00	8.39	8.05

Neil received a 10-year $130 000 loan at 6% p.a. monthly reducible interest. If the interest rate was increased to 7% p.a., find the increase in the total repayments needed to clear the debt.

32 ANZ offers the choice of two types of loans:

- Option 1: 5.2% p.a. flat rate interest over 25 years.
- Option 2: 8.2% p.a. reducible interest over 15 years.

The following financial table displays the monthly repayments on a $1000 loan:

Annual interest rate	15 years	20 years	25 years	30 years	35 years
4.2%	9.06	7.67	6.83	6.28	5.88
5.2%	9.89	8.5	7.67	7.11	6.71
6.2%	10.72	9.33	8.5	7.94	7.55
7.2%	11.56	10.17	9.33	8.78	8.38
8.2%	12.39	11	10.17	9.61	9.21
9.2%	13.22	11.83	11	10.44	10.05

a Calculate the total repayments to be made with Option 1 on a $170 000 loan.

b Calculate the total repayments to be made with Option 2 on a $170 000 loan.

c Which loan will cost less in total?

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

33 Eileen is deciding between two 20-year loans to finance the purchase of a new $\$5000$ salon.

- Option 1: 6% p.a. reducible interest along with a one-off $\$535$ loan application fee.
- Option 2: 8% p.a. reducible interest along with a monthly account fee of $\$41$.

The following financial table displays the monthly repayments on a $\$1000$ loan:

Annual interest rate	15 years	20 years	25 years	30 years	35 years
4%	7.40	6.06	5.28	4.77	4.43
5%	7.91	6.60	5.85	5.37	5.05
6%	8.44	7.16	6.44	6.00	5.70
7%	8.99	7.75	7.07	6.65	6.39
8%	9.56	8.36	7.72	7.34	7.10
9%	10.14	9.00	8.39	8.05	7.84

Which loan should she choose? Explain your answer.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647